

Modeling Pit Stop Strategy in Formula 1: A Machine Learning Approach

He Sun, Yilin Wang, Weifan Wu, Peter Ye

Cornell Tech, Cornell University

INFO 5368-030 Practical Applications in Machine Learning

Abstract—Pit stop timing is a central part of Formula 1 race strategy because it directly affects tire performance, track position, and final race outcomes. In this project, we study whether a driver will pit on the next lap using publicly available lap-level race data. We compare two machine learning methods implemented entirely in NumPy from scratch: logistic regression and a one-hidden-layer artificial neural network. We evaluate model performance using precision, recall, F1-score, ROC-AUC, and accuracy, with F1-score used as the primary metric because pit stops are the minority class. Using a temporal split with 2022 and 2024 for training and validation and 2025 as the held-out test set, the best-performing model is the threshold-tuned neural network, which achieves a test F1-score of 0.6263 and ROC-AUC of 0.7943. We also deploy the final model in a Streamlit application that allows users to input race conditions, view real-time pit-stop predictions, and explore model behavior through interactive analysis. The project shows that lap-level pit prediction is feasible using publicly available structured data and relatively small models when preprocessing, leakage control, and threshold selection are handled carefully.

I. INTRODUCTION

A. Problem and Importance

In Formula 1, race outcomes are strongly influenced by in-race strategy, and pit stop timing is one of the most important decisions. Teams must balance tire degradation, lap-time changes, race progress, traffic, and track position when deciding whether to pit. A poorly timed stop can lead to position losses that are difficult to recover, especially on tracks where overtaking is limited. For this reason, predicting pit-stop decisions is a meaningful machine learning problem and a useful strategy-analysis task.

Pit-stop decisions are also difficult to model because they depend on several factors at the same time. Some of these factors are mechanical, such

as tire age or performance loss, while others are tactical, such as current position and the behavior of nearby competitors. This makes the problem well suited for comparing simple interpretable models and slightly more flexible nonlinear models.

B. Overview of Existing Work

Traditional approaches to Formula 1 strategy rely on expert knowledge, race simulation, and predefined rules [1]. These methods can capture general patterns, but they often treat pit timing as an external decision rather than a quantity to be predicted directly from current race conditions. More recent work has explored learning-based methods for strategy modeling [2]. Other recent studies use richer sequential or telemetry-heavy settings and obtain stronger performance, but at the cost of more complex models, more detailed data, and external machine learning frameworks [3].

Although this prior work is useful, much of it is less aligned with the setting of this course project. Our project focuses on publicly available structured race data, relatively small models, and fully from-scratch implementations.

C. Gap in Existing Work

Despite the availability of public lap-level Formula 1 data, there is still limited work that studies next-lap pit decisions using small, from-scratch models in a leakage-controlled setting. Existing public work often focuses on overall race outcomes, simulator-based optimization, or richer telemetry settings. This creates a gap between publicly available structured data and practical pit-stop prediction.

D. Project Approach

In this project, we predict whether a driver will pit on the next lap. We compare logistic regression

as an interpretable baseline with a one-hidden-layer neural network that can capture nonlinear relationships. We evaluate performance using precision, recall, F1-score, ROC-AUC, and accuracy. We also build a Streamlit application that allows users to input race conditions and receive real-time pit-stop predictions.

E. Broader Impact

This project provides a data-driven perspective on race strategy using publicly available race data. It supports more systematic analysis of pit decisions and offers an accessible tool for students, fans, and analysts who want to explore Formula 1 strategy quantitatively. More broadly, the workflow used here can transfer to other event-prediction problems in sports and operations settings, where a short-term decision must be made based on evolving structured features.

II. BACKGROUND

A. Project Scope

Our scope is the next-lap pit decision problem: given the state of a driver at lap t , we predict whether the driver will pit on lap $t + 1$. We focus on this short horizon because current race-state features, such as tire life, recent lap-time behavior, and race progress, are most informative at that scale.

B. Comparisons to Prior Work

Heilmeier et al. [1] study Formula 1 strategy using simulation-based methods. Their work helps explain race outcomes under different pit sequences, but pit timing itself is treated as an external input. Heilmeier et al. [2] move closer to our setting by using neural networks for strategy decisions, but their work is based on simulator-generated data rather than public lap-level race data. Fatima and Johrendt [3] use deeper sequence models in richer telemetry settings. Compared with these studies, our project focuses on publicly available structured race data and on small from-scratch models in a strict temporal evaluation pipeline.

C. Common Knowledge and Discussion

There is ongoing discussion about how pit stop decisions are made during a race. Some approaches emphasize tire degradation, while others place

greater importance on tactical signals such as track position, traffic, and broader race context [1], [2]. In practice, these factors interact and change throughout the race. This makes pit-stop prediction difficult to model using a single rule.

D. Knowledge Gap and Contribution

Most prior public work either optimizes strategy inside a simulator, uses richer telemetry, or depends on larger models and off-the-shelf learning pipelines. Our contribution is to show that next-lap pit-stop prediction can be studied effectively using public lap-level data, strict temporal splitting, from-scratch NumPy implementations, and a deployed Streamlit application. We also provide a direct comparison between logistic regression and a one-hidden-layer neural network under the same preprocessing and evaluation pipeline.

III. END-TO-END MACHINE LEARNING PIPELINE

A. Offline Model Training and Evaluation

1) Data Collection, Exploration, and Processing: We use the `f1_strategy_dataset_v4` dataset from Kaggle, which is a processed version of publicly accessible Formula 1 timing and telemetry data collected through the FastF1 Python package [4]. The raw dataset contains 101,371 lap-level observations across the 2022, 2023, 2024, and 2025 seasons. The data includes numerical, categorical, and time-related features such as lap number, tire life, position, lap time, compound, and race information. We use the dataset to predict `PitNextLap`, which serves as the ground-truth label. Since the label is already provided in the dataset, no manual annotation is required.

During exploration, we checked missing values, target distribution, compound distribution, outliers, and feature relationships. One important finding was that the 2023 subset had an anomalously low pit rate of about 3%, while the other years were all around 32%–33%. We therefore excluded the 2023 season to reduce label noise. We also identified extreme outliers in variables such as `LapTime (s)` and `LapTime_Delta`, which motivated clipping based on train-year quantiles. In addition, a correlation analysis showed that `TyreLife` and `Cumulative_Degradation` were among

the most relevant variables for the target, while Normalized_TyreLife was highly redundant with TyreLife. The boxplot of TyreLife grouped by the target also showed that laps followed by a pit stop generally had higher tire life values.

Figure 1 shows the class distribution of PitNextLap, and Figure 2 shows the distribution of TyreLife grouped by the target label. These figures support two main observations: the task has moderate class imbalance, and tire life appears to be useful for prediction.

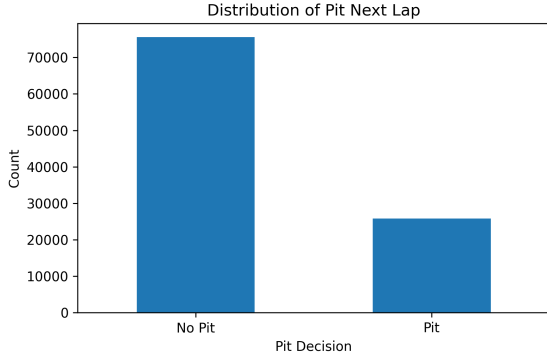


Fig. 1. Class distribution of PitNextLap. The dataset contains more non-pit laps than pit-next-lap instances, which motivates the use of precision, recall, and F1-score.

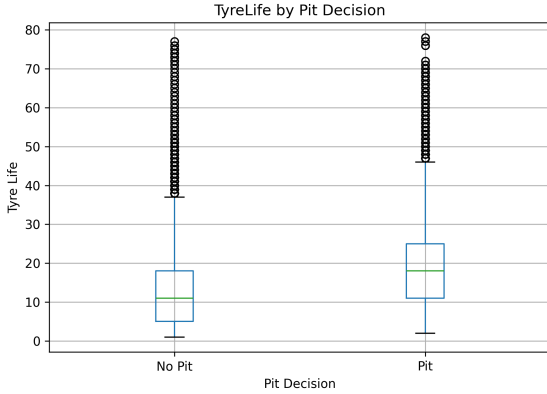


Fig. 2. Distribution of TyreLife grouped by PitNextLap. Laps followed by a pit stop generally have higher tire life values.

Our preprocessing pipeline includes removal of the 2023 season, removal of the leakage-prone PitStop column, removal of the redundant Normalized_TyreLife feature, dropping 66 rows with missing Compound, outlier clipping using train-year quantiles, feature creation through

Degradation_Rate, one-hot encoding of tire compound, label encoding of driver identity, and train-set-based standardization of numerical variables. The final feature set contains 16 inputs: 10 standardized numerical features, 5 one-hot encoded compound features, and 1 label-encoded driver feature. After preprocessing, the dataset contains zero missing values.

We use a time-based split to avoid temporal leakage. The training set consists of the 2022 and 2024 seasons with 49,351 rows, and the held-out test set consists of the 2025 season with 27,040 rows. For threshold tuning and controlled model comparison, we use Train=2022 and Val=2024. The class rate is 32.81% on the train fold and 32.67% on the test fold, so the temporal split preserves the marginal label distribution.

2) *Methods and Model Training*: We formulate the task as a binary classification problem. Let $x \in R^d$ denote the lap-level feature vector and let $y \in \{0, 1\}$ denote the target label, where $y = 1$ indicates a pit stop on the next lap.

We compare two machine learning models.

The first model is logistic regression. It serves as an interpretable baseline and outputs

$$P(y = 1 | x) = \sigma(w^\top x + b),$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

The model is trained by minimizing binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)].$$

We also evaluate an L2-regularized version of logistic regression, where the loss becomes

$$\mathcal{L}_{L2} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + \frac{\lambda}{2} \|w\|_2^2,$$

with the regularization strength λ tuned over the grid $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ on the 2024 validation fold. The selected value is $\lambda^* = 0.01$.

The second model is a one-hidden-layer artificial neural network implemented in NumPy:

$$h = \phi(W_1 x + b_1), \quad \hat{y} = \sigma(W_2 h + b_2),$$

where $\phi(\cdot)$ is the ReLU activation function and $\sigma(\cdot)$ is the sigmoid output function. The ANN uses a $16 \rightarrow 128 \rightarrow 1$ architecture and He initialization [5]. Both models use binary cross-entropy as the objective.

Both models are trained entirely from scratch in NumPy, without using machine learning libraries for model training. Logistic regression is well suited to this task because it provides a simple and interpretable baseline. The neural network is well suited because it can capture nonlinear interactions among features such as tire life, race progress, and degradation behavior. The input to both models is the same 16-feature vector, and the output is a probability of next-lap pit behavior.

3) *Model Evaluation*: We evaluate model performance using precision, recall, F1-score, ROC-AUC, and accuracy. Precision, recall, and F1-score are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

These metrics are appropriate because pit-stop events are less frequent than non-pit laps. Accuracy alone would be misleading, while precision, recall, and F1-score better describe performance on the positive class. We use F1-score as the primary metric because it balances false positives and false negatives. ROC-AUC is included to measure ranking quality across thresholds.

Our experimental design follows a strict temporal protocol. We train on 2022 and validate on 2024 for hyperparameter and threshold tuning, then retrain on 2022+2024 and evaluate on the held-out 2025 test set. This setup prevents leakage from future seasons into the training process. Outlier clipping boundaries and standardization statistics are also computed from train-year rows only.

We perform threshold tuning using the validation fold. Instead of fixing the decision threshold at 0.5, we select thresholds that maximize F1-score on the 2024 validation set. The selected thresholds are approximately 0.2424 for logistic regression and 0.2418 for the ANN. We also tune the L2

regularization strength for logistic regression on the same validation fold.

To reduce overfitting, we use a temporal hold-out, keep the ANN architecture relatively small, and compare train and validation behavior during training. The L2 regularization experiment for logistic regression is also used to evaluate whether the linear model benefits from additional weight control.

4) *Results*: Table I summarizes the main test-set results on the 2025 season. The best-performing model is the threshold-tuned ANN, which achieves an F1-score of 0.6263 and ROC-AUC of 0.7943. The worst-performing model is logistic regression under the default threshold of 0.5, with an F1-score of 0.3360.

The neural network outperforms logistic regression under both the default and tuned thresholds. Threshold tuning substantially improves recall for logistic regression and moderately improves F1-score for the ANN. For the ANN, threshold tuning increases recall from 0.6334 to 0.8825 and increases F1-score from 0.6179 to 0.6263. This suggests that the ANN already learns useful ranking information under the default threshold, but benefits from a decision rule that is better aligned with the class distribution.

Figure 3 shows the ROC curves of both models on the 2025 test set. The ANN dominates the logistic regression baseline at every operating point, and the gap is largest in the high-recall region, which is the region that matters most for a minority-positive task.

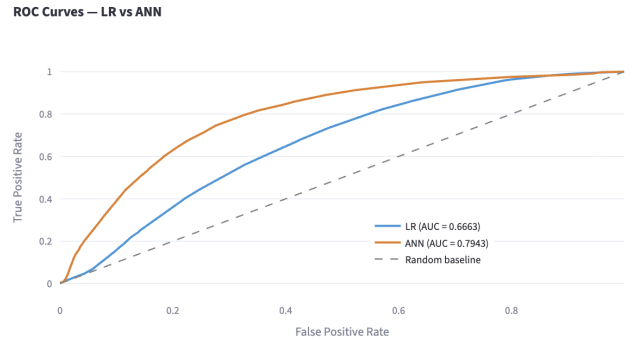


Fig. 3. ROC curves on the 2025 test set. The ANN reaches an AUC of 0.7943, while the logistic regression baseline reaches 0.6663. The ANN dominates the logistic regression baseline across all operating points.

We also evaluated L2 regularization for logistic regression. The tuned L2 model performs slightly

TABLE I
TEST-SET (2025) PERFORMANCE. BEST PER METRIC IN BOLD; DEPLOYED MODEL MARKED WITH *.

Model	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC
LR (default $t=0.5$)	0.5000	0.6596	0.4630	0.2637	0.3360	0.6663
LR tuned, no reg	0.2424	0.5966	0.4301	0.7218	0.5390	0.6663
LR tuned, L2 ($\lambda=0.01$)	0.2568	0.5935	0.4268	0.7128	0.5339	0.6643
ANN (default $t=0.5$)	0.5000	0.7441	0.6032	0.6334	0.6179	0.7943
ANN tuned *	0.2418	0.6559	0.4853	0.8825	0.6263	0.7943

worse than the non-regularized tuned logistic regression on the test set. This suggests that the main limitation of the linear model is not overfitting but model capacity. In contrast, the ANN benefits from its ability to capture nonlinear feature interactions.

For interpretability, we examined logistic-regression coefficients. The largest-magnitude features include TyreLife, Stint, RaceProgress, and compound indicators, which is consistent with the idea that pit decisions are strongly related to tire condition and race progression. Tactical variables such as Position and Position_Change have smaller coefficient magnitudes, which suggests that condition-based signals are more important than tactical ones in this feature set. Figure 4 visualizes the standardized coefficients sorted by magnitude.

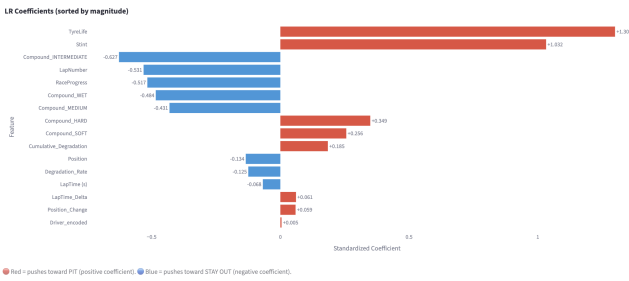


Fig. 4. Standardized logistic-regression coefficients sorted by magnitude. TyreLife (+1.300) and Stint (+1.032) are the two dominant positive predictors. Compound indicators and LapNumber carry the next-largest weights, while tactical variables such as Position and Position_Change have visibly smaller coefficients.

5) *Model Deployment and Front-End Application Using Streamlit*: We deploy the selected model as a Streamlit application. The deployed model is the threshold-tuned one-hidden-layer ANN because it provides the strongest F1-score and ROC-AUC while still remaining small and efficient. Although

logistic regression is more interpretable, the ANN offers better predictive performance, and we retain logistic-regression coefficient analysis as a complementary interpretation tool.

The application is designed for students, Formula 1 fans, and strategy-oriented users who want to explore pit-stop decisions under different race conditions. The user interface includes three main pages: a prediction page, a sensitivity analysis page, and a model performance page.

The prediction page allows users to enter race conditions such as driver, tire compound, lap number, stint, tire life, position, race progress, lap time, lap-time delta, cumulative degradation, position change, and degradation rate. The app then returns a pit-stop probability and a binary prediction using the tuned threshold. The sensitivity analysis page allows the user to vary one feature while holding others fixed in order to observe how the predicted probability changes. The model performance page compares the logistic regression and neural network models using stored evaluation outputs.

As shown in Figure 5, the application uses a simple layout with an input panel, a prediction output area, and analysis panels. On the back end, the app reads the saved model weights, feature order, scaling parameters, driver mapping, and model configuration. User inputs are transformed into the same feature format used during training before the model generates a real-time prediction. This design allows the deployed application to stay consistent with the offline training pipeline.

6) *Limitations and Ethical Considerations*: Although the threshold-tuned ANN reaches an F1-score of 0.6263 on the 2025 hold-out, several limitations remain. First, our feature set is restricted to lap-level structured data and does not include

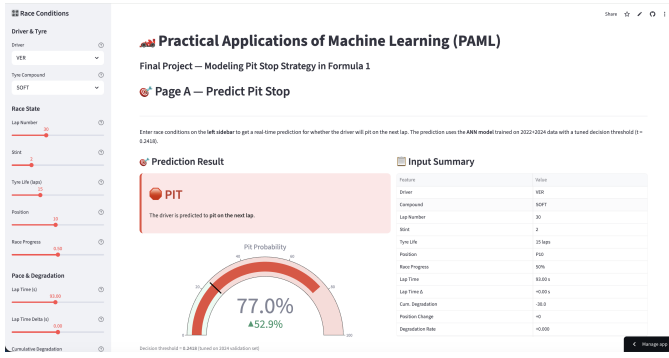


Fig. 5. Streamlit interface for interactive pit-stop prediction. The layout includes an input panel, a prediction output area, and interactive analysis panels.

detailed telemetry such as brake, throttle, or tire-temperature signals, which are typically available to professional teams and are used by deeper sequence models in prior work [3]. Second, the dataset does not contain track identity or weather information, so the model cannot learn track-specific pit windows or weather-driven decisions. Third, we use only one held-out season for testing, which limits how well our results generalize across multiple seasons with different regulations or tire compounds. Fourth, the driver feature is encoded as an integer label rather than a learned embedding, which may introduce a small amount of unintended ordinal structure into the linear model.

We also considered several ethical aspects. The deployed application is intended as a decision-support and educational tool rather than an autonomous strategy system, so the final pit decision still belongs to the race engineer. Because the application is publicly accessible, it could in principle be used in sports-betting contexts, but the current F1-score is well below what such use cases would require, and we do not present the application as a betting tool. All data used in this project comes from publicly available FastF1 timing feeds [4], so no proprietary or confidential team data is involved. Finally, because driver identity is one of the model inputs, predictions are conditioned on individual drivers, and we did not observe systematic accuracy gaps across drivers in our evaluation. A fairness audit at the per-driver level would still be appropriate before any high-stakes deployment.

7) *Conclusion:* The goal of this project was to predict next-lap pit decisions in Formula 1 using publicly available structured race data. We developed two from-scratch NumPy models, logistic regression and a one-hidden-layer neural network, and evaluated them using a strict temporal protocol. The threshold-tuned ANN achieved the best overall performance on the held-out 2025 season.

For our target users, the main value of this project is that it provides an accessible and interactive way to explore race strategy without requiring proprietary telemetry or specialized analytics tools. More broadly, the project shows that publicly available structured motorsport data can support meaningful event-prediction tasks when preprocessing, leakage control, and threshold tuning are handled carefully.

In future work, we would like to explore richer sequential models, cost-sensitive learning, and additional contextual features such as track characteristics and weather conditions.

IV. TEAM MEMBER CONTRIBUTION

Technical Contributions: He Sun was responsible for data preprocessing and feature engineering, including leakage-aware clipping and preprocessing design. Yilin Wang implemented the logistic regression model, including the L2 regularization experiment, and supported evaluation and tuning. Weifan Wu implemented the one-hidden-layer neural network, conducted threshold tuning, and developed and deployed the Streamlit application. Peter Ye contributed exploratory data analysis, figure preparation, and result organization.

Writing Contributions: He Sun drafted the Background section. Yilin Wang wrote the Methods and Evaluation sections. Weifan Wu wrote the Streamlit and Conclusion sections. Peter Ye wrote the Abstract, Introduction, and Results discussion and reviewed the report for consistency.

V. CODE, DOCUMENTATION, AND DEPLOYMENT

The GitHub repository for this project is available at: <https://github.com/Cornell-Tech-PAML-Course-2026/final-project-submission-ww.git>
<https://github.com/ww566-wwf/INFO-5368-PAML-Project-Team-ORDS.git>

The repository contains the Jupyter notebooks used for data preprocessing and model training, the Streamlit application code, the dataset used for training and evaluation, and a README file with instructions for running the application. It also includes the saved artifacts required for deployment, including model weights, feature-order information, scaling parameters, driver mapping, and model configuration files.

The deployed Streamlit application is available at: <https://ww566-wwf-info-5368-paml-project-team-ords-app-wpj3c.streamlit.app>

REFERENCES

- [1] A. Heilmeier, M. Graf, and M. Lienkamp, "A race simulation for strategy decisions in circuit motorsports," in *Proceedings of the 21st IEEE International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2018, pp. 2986–2993.
- [2] A. Heilmeier, A. Thomaser, M. Graf, and J. Betz, "Virtual strategy engineer: Using artificial neural networks for making race strategy decisions in circuit motorsport," *Applied Sciences*, vol. 10, no. 21, p. 7805, 2020.
- [3] M. Fatima and J. Johrendt, "Data-driven pit stop decision support for formula 1 using deep learning models," *Frontiers in Artificial Intelligence*, vol. 8, p. 1673148, 2025.
- [4] P. Schäfer, "FastF1: A python package for accessing and analyzing formula 1 data," <https://github.com/theOehrly/Fast-F1>, 2020, accessed: 2025.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proceedings of the 2015 IEEE International Conference on Computer Vision (ICCV)*. IEEE, 2015, pp. 1026–1034.